



# MarketingAI

AI-Powered Social Media Manager

Complete Platform Documentation with Screenshots

March 2026 · Next.js · Firebase · Google Gemini AI

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## 1. Platform Overview & Tech Stack

MarketingAI is a full-stack SaaS marketing platform that helps businesses generate AI-written content, schedule it, publish it to Instagram and LinkedIn, analyze its performance, and learn from data using an intelligent Strategy Agent — all in one place.

Core Loop:

Post → Publish → Collect Engagement Data → Analyze → Decide → Post Better

Every piece of the system feeds this loop. The Strategy Agent closes it automatically.

Technology Stack

| Layer       | Technology                               | Purpose                                            |
|-------------|------------------------------------------|----------------------------------------------------|
| Frontend    | Next.js 14 (App Router) + TypeScript     | Full-stack React framework, routing, SSR           |
| Styling     | Tailwind CSS + Framer Motion             | UI design + page transition animations             |
| Auth        | Firebase Auth + Admin SDK                | JWT-based login, token verification per request    |
| Database    | Firestore (Firebase)                     | Posts, users, insights, agent strategies           |
| Primary AI  | Google Gemini 2.0 Flash                  | Captions, sentiment, strategy, replies, tone guard |
| AI Fallback | Hugging Face Inference API               | Backup if Gemini fails                             |
| Social APIs | Facebook Graph API v18 + LinkedIn API v2 | Publishing, metrics, comments, replies             |
| Images      | Pexels API                               | Free stock photo search via server-side proxy      |

2. Authentication System

Firebase Auth handles all authentication. Every API route is protected by verifying the user's JWT token before processing any request.

Login Flow — Step by Step

- 1 User enters email + password on `/login`
- 2 Firebase Auth SDK calls `signInWithEmailAndPassword()`
- 3 Firebase issues a short-lived JWT ID Token (1 hour TTL, auto-refreshed)
- 4 Every API call sends: `Authorization: Bearer <token>`
- 5 Each API route calls `getAuth().verifyIdToken(token)` — rejects any expired or tampered tokens

## Security Principles

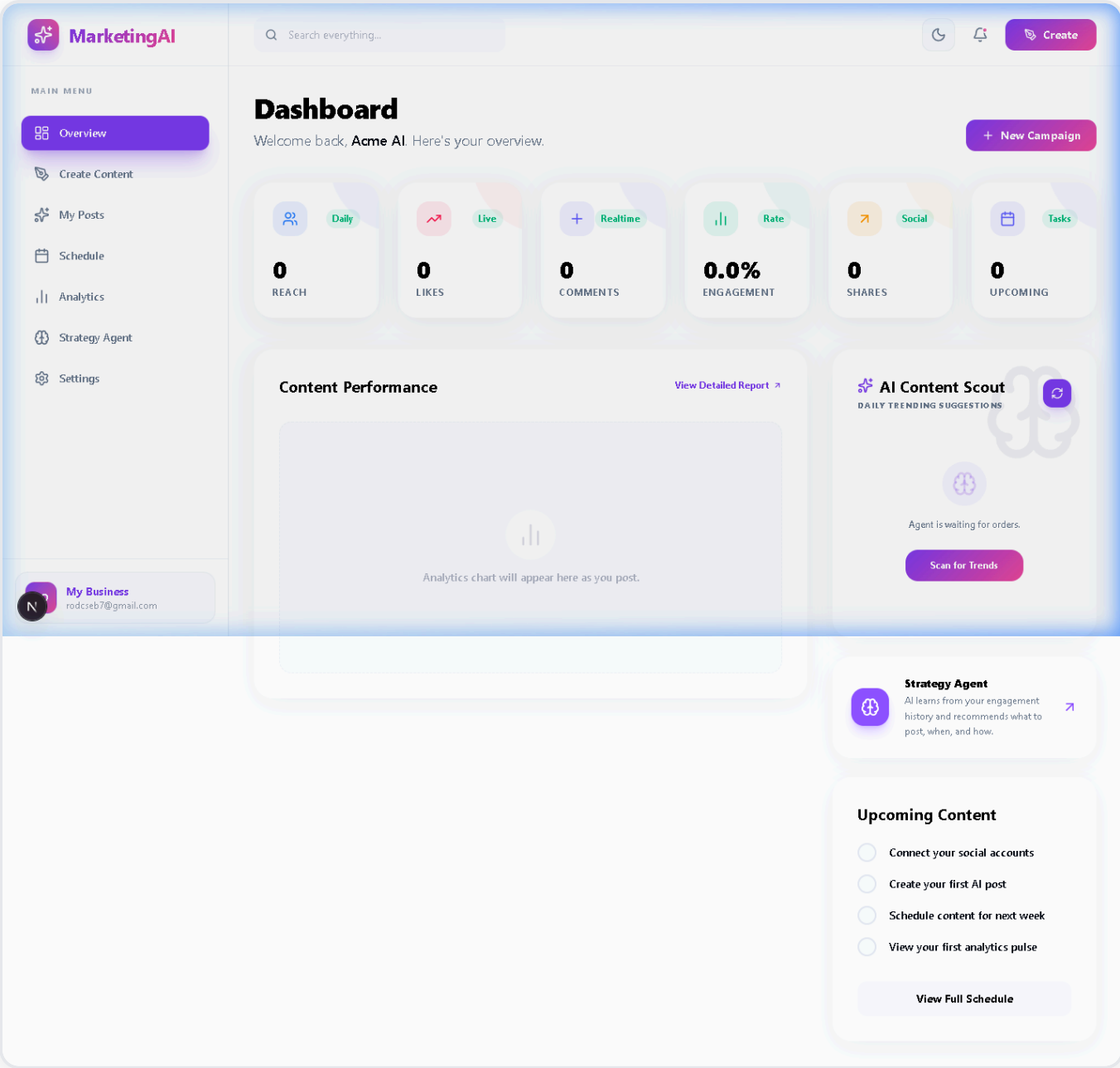
- Tokens stored in React memory, never in `localStorage`
- Dashboard layout auto-redirects to `/login` if no valid session
- Every API verifies `post.userId === uid` before modifying any data (ownership check)
- Social access tokens (Instagram, LinkedIn) stored server-side in Firestore — never exposed to frontend

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## 3. Dashboard Home (`/dashboard`)

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The central hub showing a real-time overview of performance, AI suggestions, and quick access to all features.



Dashboard Home — Stats row, AI Content Scout panel, Strategy Agent banner, and Upcoming Content checklist

Page Elements Explained

1 HEADER STATS ROW (6 CARDS)

| Card            | Data Source                | Formula / Logic                         |
|-----------------|----------------------------|-----------------------------------------|
| Reach           | Firestore metrics.reach    | Sum of reach across all published posts |
| Likes           | Firestore metrics.likes    | Sum of all likes                        |
| Comments        | Firestore metrics.comments | Sum of all comments                     |
| Engagement Rate | Computed on load           | (likes + comments) / reach × 100        |

|          |                          |                                          |
|----------|--------------------------|------------------------------------------|
| Shares   | Firestore metrics.shares | Sum of all shares                        |
| Upcoming | Firestore query          | Count of posts with status = "scheduled" |

## ② AI CONTENT SCOUT PANEL

Clicking "Scan for Trends" calls `POST /api/agent/scout`. Gemini generates 3 trending topic posts tailored to the user's business profile and saves them as suggested drafts in Firestore. Posts appear immediately in this panel for review.

## ③ STRATEGY AGENT BANNER

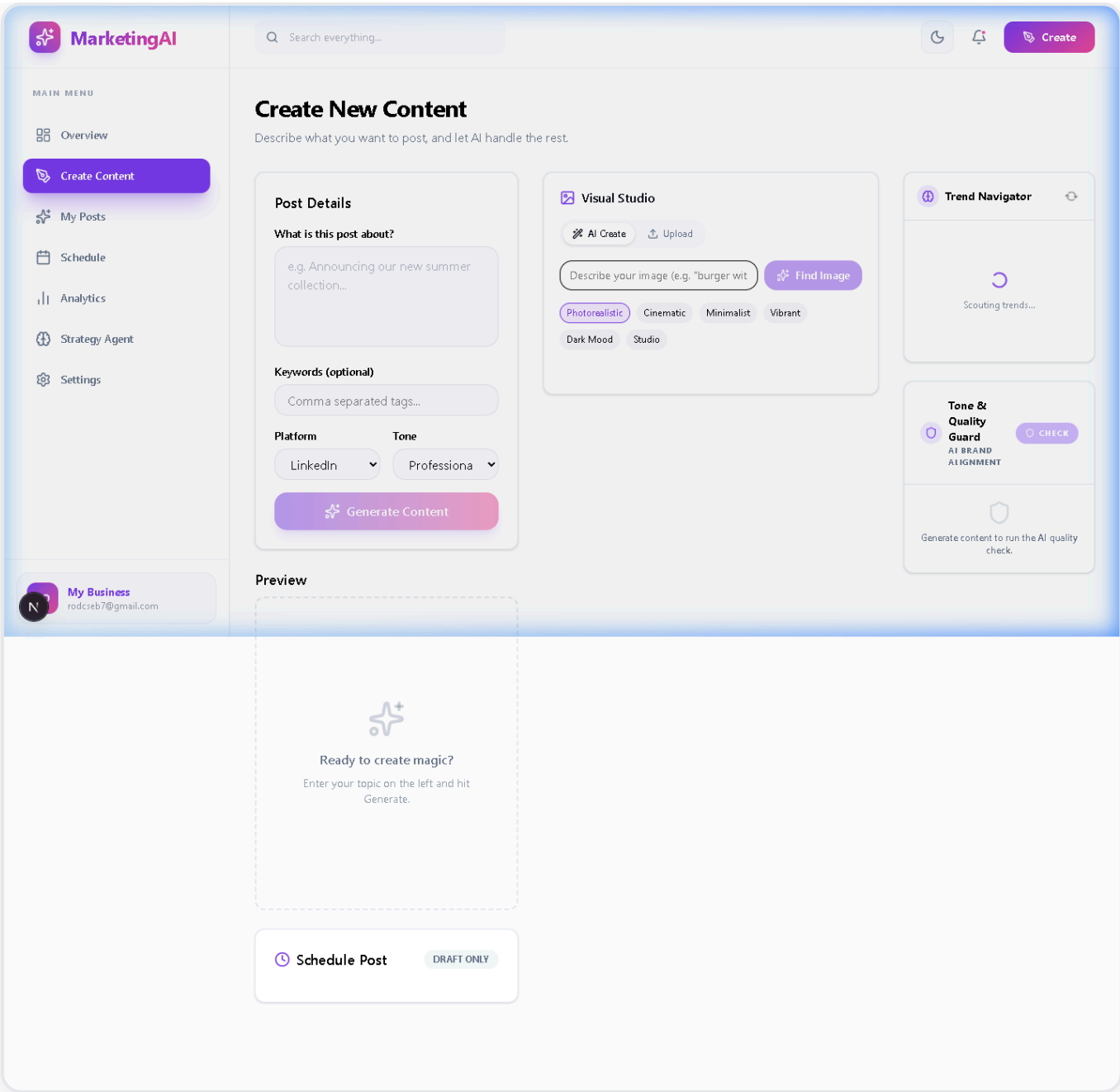
A click-through card that links to `/dashboard/strategy`. Describes the agent's role: it learns from engagement history and recommends what to post, when, and how.

## ④ UPCOMING CONTENT

Lists upcoming scheduled posts. For new users, shows a "getting started" checklist: connect accounts → create first post → schedule → view analytics.

# 4. Create Content Page (`/dashboard/create`)

The AI-powered content creation workspace. Enter a topic, pick a tone and platform, and Gemini writes a publish-ready caption with hashtags.



Create Content — Post Details form (left), Visual Studio (center), Trend Navigator + Tone Guard (right)

## Left Panel — Post Details Form

| Field                    | Purpose                                                                     |
|--------------------------|-----------------------------------------------------------------------------|
| What is this post about? | Main topic/brief — becomes the core Gemini prompt                           |
| Keywords (optional)      | Extra terms to weave into the caption                                       |
| Platform                 | Instagram / LinkedIn / Twitter — adjusts character limits and writing style |
| Tone                     | Professional / Casual / Humorous / Inspirational / Persuasive               |
| Generate Content         | Calls POST /api/content/generate → returns Gemini-written caption           |

## Gemini Prompt Structure

```
You are a {tone} social media expert for a {industry} business.  
Write a {platform} post about: {topic}  
Keywords to include: {keywords}  
Include 5 relevant hashtags. Keep it under {character_limit} characters.
```

## Center Panel — Visual Studio

- **AI Create** — generates an image using Gemini's image generation capability based on the post topic
- **Find Image** — searches Pexels API (routed through server-side proxy) for free stock photos
- **Upload** — user uploads their own image

Images are stored as CDN URLs (never uploaded to Firebase Storage) to eliminate storage costs.

## Right Sidebars

### TREND NAVIGATOR

Auto-generates 5 trending topics for the user's industry using Gemini with the current date as context. Refreshes each page load. Click a topic to auto-fill the post form.

### TONE & QUALITY GUARD

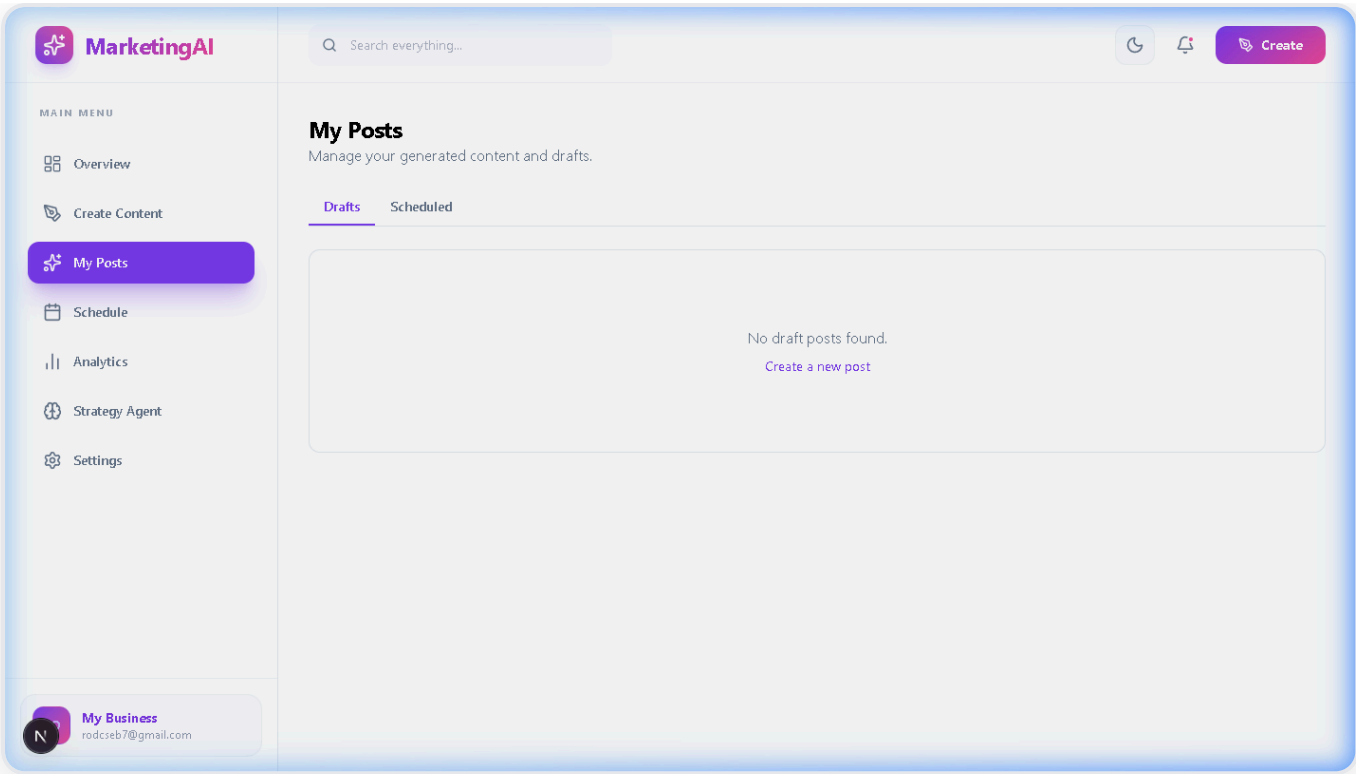
After generating content, "CHECK" calls POST /api/agent/tone-guard. Gemini compares the writing style to the user's configured brand tone. If misaligned (e.g., too formal for a casual brand), it highlights the issue with a rewrite suggestion.

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## 5. My Posts Page (/dashboard/posts)

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Central management hub for all posts regardless of status.



My Posts — All content in a unified view with status badges and quick actions

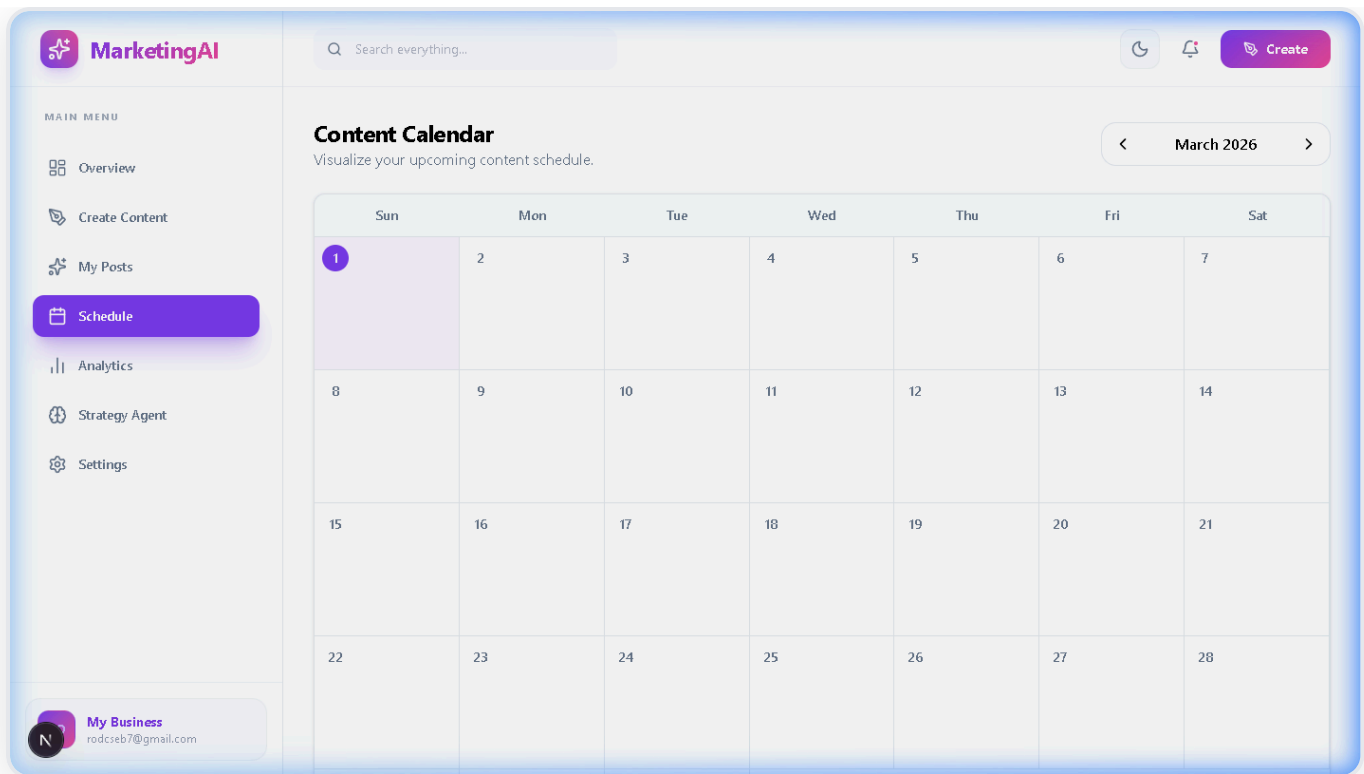
Post Statuses

| Status    | Meaning                      | Available Actions                                  |
|-----------|------------------------------|----------------------------------------------------|
| DRAFT     | Saved but not yet scheduled  | Schedule, Publish Now, Edit, Delete                |
| SUGGESTED | Created by AI Content Scout  | Review, Customize, then Schedule or Publish        |
| SCHEDULED | Queued for future publishing | Edit time, Cancel, Publish Now                     |
| PUBLISHED | Live on the platform         | View metrics, Analyze Sentiment, Reply to comments |

6. Schedule / Calendar (/dashboard/schedule)

Month-at-a-glance view of all scheduled and published content to help visualize posting consistency and gaps.





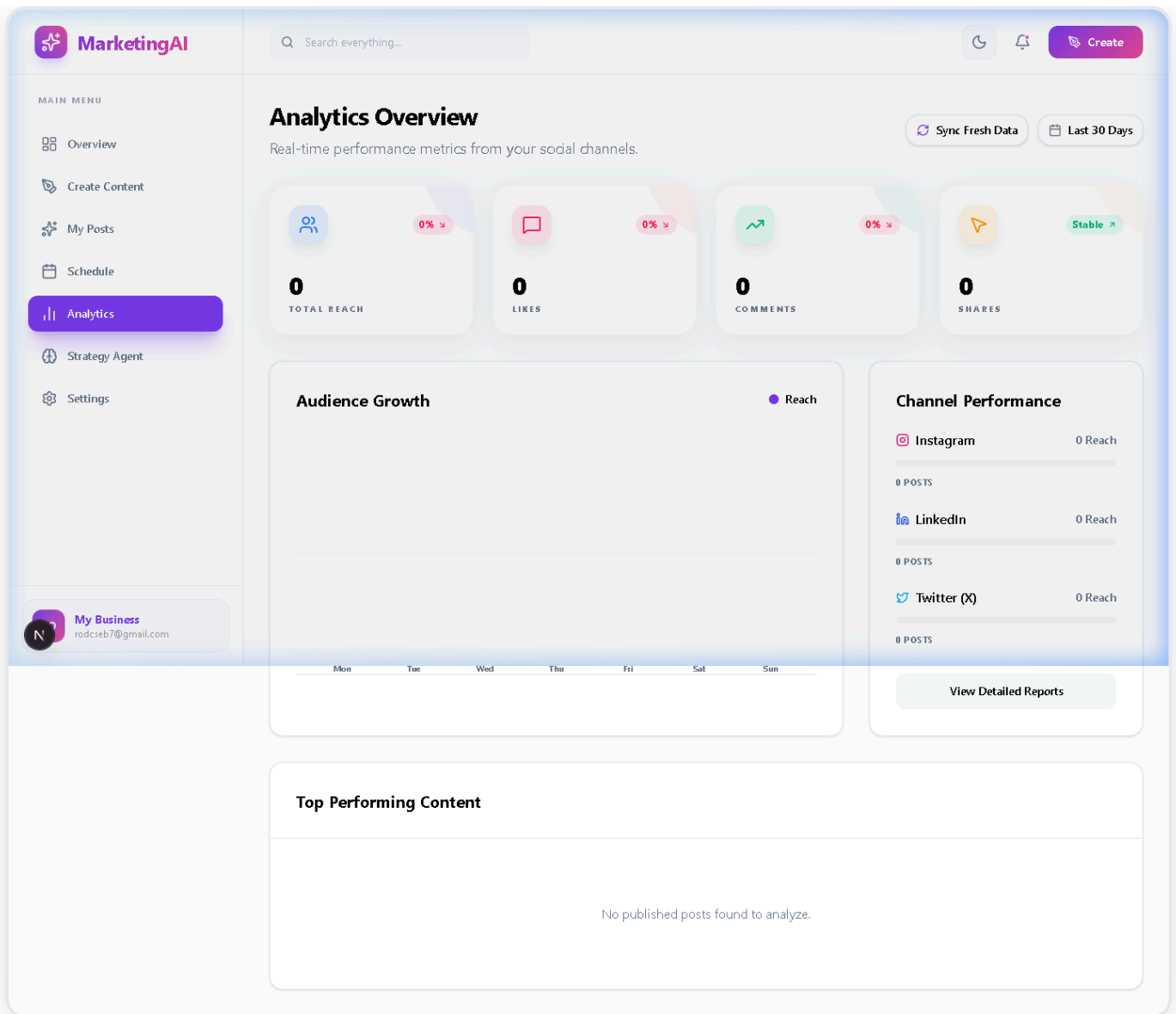
*Schedule — Monthly calendar with posts shown as color-coded chips on their scheduled dates*

## How It Works

- Built with pure React state — no external calendar library
- Each day cell queries Firestore for posts whose `scheduledAt` falls on that date
- Posts appear as colored chips (blue = scheduled, green = published)
- Click any chip to open that post for editing
- Seeing gaps in the calendar motivates consistent posting

## 7. Analytics Page (/dashboard/analytics)

Full engagement analytics with real-time metrics sync from Instagram and LinkedIn, plus AI-powered sentiment analysis on comments.



Analytics — Total Reach, Likes, Comments, Shares, Audience Growth chart, Channel Performance breakdown

## Sync Fresh Data — How It Works

- 1 User clicks "Sync Fresh Data" → calls POST `/api/analytics/sync`
- 2 Fetches all published posts for the user from Firestore
- 3 For each post, calls platform API: Instagram Graph API `/media/{id}/insights` or LinkedIn `/v2/socialMetadata/{id}`
- 4 Writes back metrics: `{ likes, comments, shares, reach }` to each Firestore post document
- 5 **Auto-triggers Observer agent in background** to re-score engagement (fire-and-forget, non-blocking)

## Analyze Pulse — Sentiment Modal

Clicking "Analyze Pulse" on any published post opens a modal:

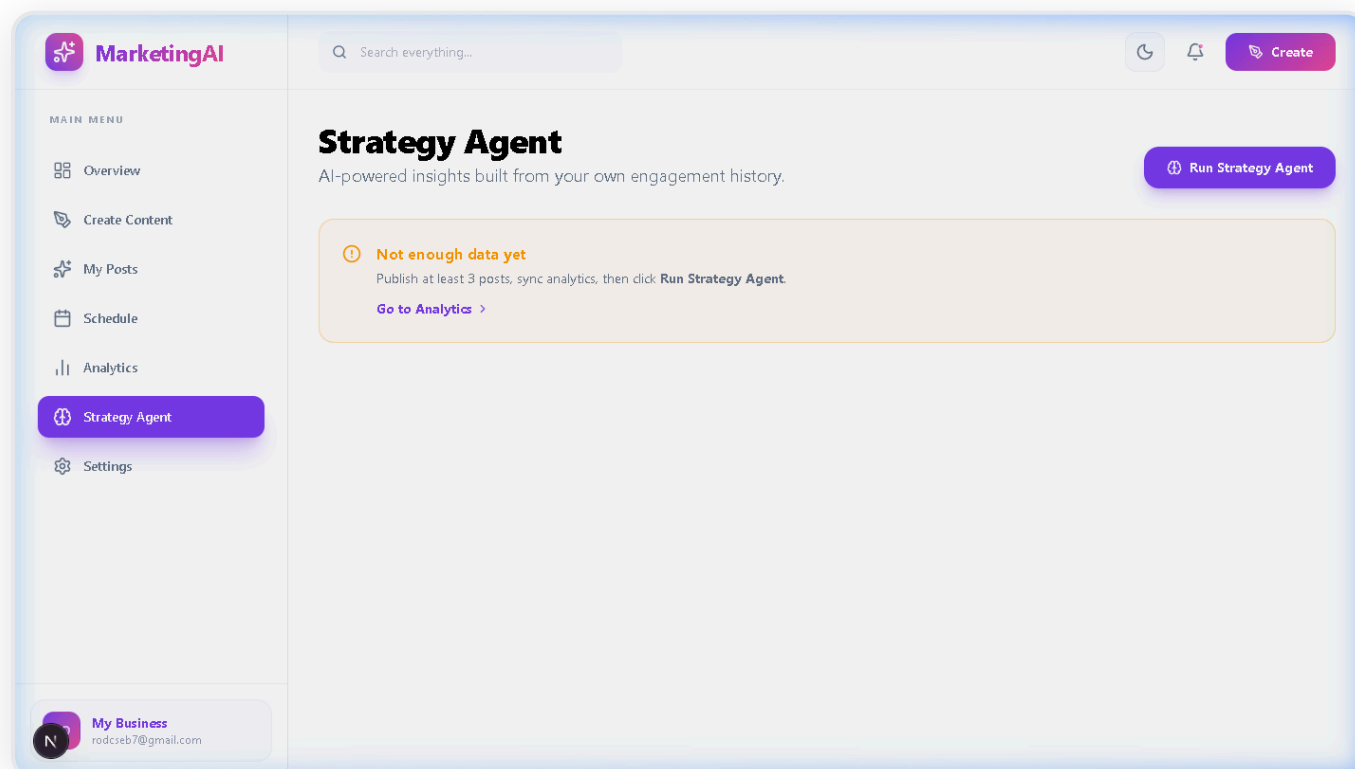
- Fetches all comments from the platform via the comments API
- Sends comment texts to Gemini: *"Analyze sentiment. Return: score 0–100, dominant emotion, top positive/negative themes, summary."*
- Displays: score gauge, positive/negative/neutral split, theme chips, individual comments

## AI Reply Suggestions (inside Pulse Modal)

- "Suggest Reply" → POST /api/agent/suggest-reply → Gemini writes brand-aligned reply
- Reply appears in an **editable textarea** — user can modify before posting
- "Post Reply" → POST /api/agent/post-reply → posts directly to Instagram (Graph API /replies) or LinkedIn (nested comments)
- After posting: textarea disabled, shows "Posted!" — prevents double-posting

# 8. Strategy Agent — Complete Deep Dive (/dashboard/strategy)

The most sophisticated feature. An analytical AI feedback loop that observes what works in your posts, finds patterns, and prescribes a data-driven weekly content strategy.



*Strategy Agent — Empty state before first run. Shows the "Run Strategy Agent" button and guidance to publish 3+ posts first.*

Agent runs in 3 sequential stages:

STAGE 1 — OBSERVE: Score every post with engagement metrics

STAGE 2 — ANALYZE: Detect patterns across hours, tones, platforms, categories

STAGE 3 — PLAN: Generate a weekly strategy with plain-English reasoning

## Stage 1 — OBSERVER MODULE

**File:** lib/agent/observer.ts | **Trigger:** POST /api/agent/observe | **Auto-runs:** After every analytics sync

### ENGAGEMENT SCORE FORMULA

```
engagementScore = (0.5 × likes) + (0.3 × comments) + (0.2 × shares)
```

Example: 100 likes, 30 comments, 10 shares

```
= (0.5 × 100) + (0.3 × 30) + (0.2 × 10)
= 50 + 9 + 2 = 61.0
```

| Action   | Weight | Reason                                                                          |
|----------|--------|---------------------------------------------------------------------------------|
| Likes    | 0.5    | Most common action, moderate signal                                             |
| Comments | 0.3    | Higher intent — active engagement is harder to get                              |
| Shares   | 0.2    | Strongest virality signal but rarest — lower weight prevents outlier distortion |

### ENGAGEMENT RATE FORMULA

```
engagementRate = (engagementScore / reach) × 100
```

Why: Normalizes performance across posts of different audience sizes.

Post A: score 50 from 500 reach = 10% → outperforms

Post B: score 200 from 5000 reach = 4%

### OTHER FIELDS COMPUTED PER POST

- postHour = publishedDate.getHours() (0–23)
- postDayOfWeek = publishedDate.getDay() (0=Sun, 6=Sat)
- hashtags = all #tags extracted via regex /[w]+/g
- contentCategory = promotional / educational / engagement / news (keyword heuristic)

## Stage 2 — ANALYZER MODULE

**File:** lib/agent/analyzer.ts | **Output:** agent\_insights/{userId} in Firestore

### PATTERN DETECTION — GROUP-BY-AVERAGE

```
For each dimension (hour, tone, platform, category):
  Group all posts by value → calculate mean(engagementScore) per group → find max

Example hourly breakdown:
Hour 07 → avg 12.3 (2 posts)
Hour 12 → avg 18.7 (3 posts)
Hour 19 → avg 41.2 (5 posts) ← bestHour = 19
Hour 21 → avg 22.1 (2 posts)
```

### 7-DAY TREND DETECTION

```
last7Avg = mean(scores for posts in last 7 days)
prev7Avg = mean(scores for posts 8-14 days ago)
delta = ((last7Avg - prev7Avg) / prev7Avg) × 100

if delta >= 5%: recentTrend = "up" ↑
if delta <= -5%: recentTrend = "down" ↓
else: recentTrend = "stable" →
```

### CONFIDENCE SCORE

| Posts Analyzed | Confidence | Label                      |
|----------------|------------|----------------------------|
| < 3            | 10%        | Very Low — not enough data |
| 3–4            | 30%        | Low                        |
| 5–9            | 60%        | Medium                     |
| 10–19          | 80%        | High                       |
| 20+            | 100%       | Very High                  |

## Stage 3 — PLANNER MODULE

**File:** lib/agent/planner.ts | **Output:** agent\_strategy/{userId} in Firestore

WEEKLY POST TARGET



```
confidence >= 80 → weeklyPostTarget = 5
confidence >= 50 → weeklyPostTarget = 4
else → weeklyPostTarget = 3
```

RECOMMENDED POST TIMES



```
Slot 1: bestDay at bestHour:00
Slot 2: bestDay + 2 days at bestHour:00
Slot 3: bestDay + 4 days at bestHour:00

If bestDay=Tuesday, bestHour=19 → ["Tue 19:00", "Thu 19:00", "Sat 19:00"]
```

CONTENT MIX FORMULA

| Best Performing Category | Promotional | Educational | Engagement | News |
|--------------------------|-------------|-------------|------------|------|
| promotional              | 50%         | 20%         | 20%        | 10%  |
| educational              | 20%         | 50%         | 20%        | 10%  |
| engagement               | 20%         | 20%         | 50%        | 10%  |
| news                     | 20%         | 30%         | 20%        | 30%  |

SAMPLE AGENT REASONING OUTPUT



```
"Posts at 19:00 have your highest average engagement score."
"Your 'Casual' tone outperforms others by ~42% in engagement."
"Instagram gives ~120% better engagement than other platforms."
"'promotional' posts drive the most engagement in your feed."
"Your engagement is trending upward — maintain current strategy momentum."
```

Dashboard UI — Every Element Explained

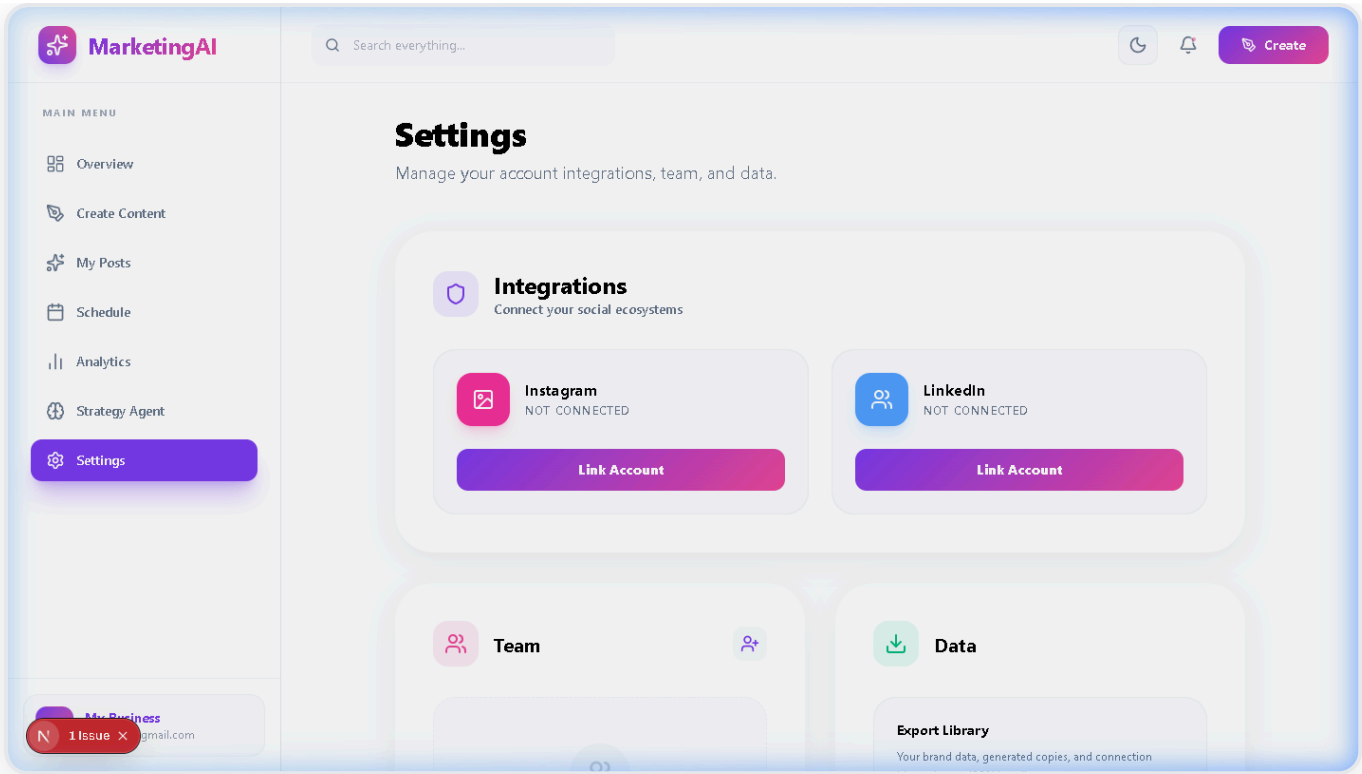
| Element                   | What It Shows                              | Why It's There                                |
|---------------------------|--------------------------------------------|-----------------------------------------------|
| Run Strategy Agent button | Triggers observe → analyze → plan pipeline | Manual control — user decides when to refresh |

|                           |                                                    |                                                   |
|---------------------------|----------------------------------------------------|---------------------------------------------------|
| Posts Analyzed card       | Total posts scored by Observer                     | Shows agent data quality                          |
| Avg Engagement Score card | Mean score across all posts                        | Baseline performance benchmark                    |
| Best Platform card        | Instagram or LinkedIn — highest avg score          | Where to focus effort                             |
| Recent Trend card         | ↑ Up / ↓ Down / → Stable (7-day delta)             | Early warning system for performance changes      |
| Hourly Heatmap            | 24-cell grid, purple opacity = avg score           | Instantly shows the best hour to post             |
| Platform Bars             | Horizontal score bars per platform                 | ROI comparison between channels                   |
| Tone & Category Bars      | Which writing style / content type wins            | Informs how to write future posts                 |
| Top Hashtags              | Tags ranked by avg post score                      | Tells user which tags drive the most engagement   |
| Weekly Strategy Card      | Post target, times, tone, platform, mix, hashtags  | Complete action plan for the week                 |
| Confidence Badge          | High / Medium / Low + %                            | Prevents over-reliance on sparse data             |
| Agent Reasoning list      | Plain-English explanation for every recommendation | Builds trust — every suggestion has a data reason |

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## 9. Settings Page (/dashboard/settings)

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Settings — Integrations tab showing Instagram and LinkedIn connection cards with "Link Account" OAuth buttons

## Instagram Connection — OAuth Flow

- 1 "Link Account" → redirects to Meta OAuth via GET /api/auth/instagram
- 2 User approves on Meta consent screen (permissions: instagram\_manage\_comments, instagram\_content\_publish)
- 3 Meta redirects to /api/auth/instagram/callback with an authorization code
- 4 Server exchanges code for long-lived access token and saves to Firestore: users/{uid}/connections/instagram

## LinkedIn Connection

Same OAuth flow via GET /api/auth/linkedin. Stores { linkedinId, accessToken, memberId } in Firestore.

# 10. Backend Architecture & API Routes

| Method   | Route             | Description                      |
|----------|-------------------|----------------------------------|
| GET/POST | /api/user/profile | Fetch or update business profile |



|                  |                              |                                                     |
|------------------|------------------------------|-----------------------------------------------------|
| GET/POST         | /api/posts                   | List all posts / create new post                    |
| GET/PATCH/DELETE | /api/posts/{id}              | Read / update / delete a single post                |
| GET              | /api/posts/{id}/comments     | Fetch comments from the social platform             |
| POST             | /api/content/generate        | Generate AI caption via Gemini                      |
| GET              | /api/auth/instagram          | Start Instagram OAuth redirect                      |
| GET              | /api/auth/instagram/callback | Complete token exchange + save                      |
| GET              | /api/auth/linkedin           | Start LinkedIn OAuth redirect                       |
| GET              | /api/auth/linkedin/callback  | Complete LinkedIn token exchange                    |
| POST             | /api/analytics/sync          | Sync metrics from platforms + auto-trigger Observer |
| POST             | /api/analytics/sentiment     | Run Gemini sentiment analysis on comments           |
| GET              | /api/analytics/timing        | Best posting time analysis                          |
| POST             | /api/agent/observe           | Run Observer module manually                        |
| POST             | /api/agent/analyze           | Run Analyzer → write agent_insights                 |
| GET/POST         | /api/agent/plan              | Fetch existing / generate new weekly strategy       |
| POST             | /api/agent/scout             | Run Content Scout — generate suggested posts        |
| POST             | /api/agent/suggest-reply     | Generate AI reply to a comment                      |
| POST             | /api/agent/post-reply        | Post a reply directly to the platform               |
| POST             | /api/agent/tone-guard        | Validate content against brand tone                 |

## 11. What's Next

### Phase 7 — Team Workspaces (Not Yet Built)

| Feature                        | What It Enables                                |
|--------------------------------|------------------------------------------------|
| Workspace Firestore collection | Group users + posts under one workspace        |
| Membership system              | Link multiple users to the same workspace      |
| Workspace-scoped posts         | Posts belong to workspace, not individual user |
| Invitation by email            | Invite team members with a magic link          |
| Role-Based Access Control      | Admin: publish   Member: draft only            |

Optional Future Enhancements

| Feature                   | Value                                                     |
|---------------------------|-----------------------------------------------------------|
| Strategy Executor         | Agent auto-schedules posts based on the strategy plan     |
| Twitter/X Integration     | Third platform connector                                  |
| Engagement Prediction     | Score a draft before publishing using historical patterns |
| Monthly Campaign Planning | Full month strategy, not just one week                    |
| Push Notifications        | Alert when posts perform unusually well or drop           |
| PDF Export                | Download performance reports as formatted PDFs            |